The Market: Demand, Supply and Equilibrium

Reference: Intermediate Microeconomics by Hal R. Varian

Indraprastha Institute of Information Technology, Delhi

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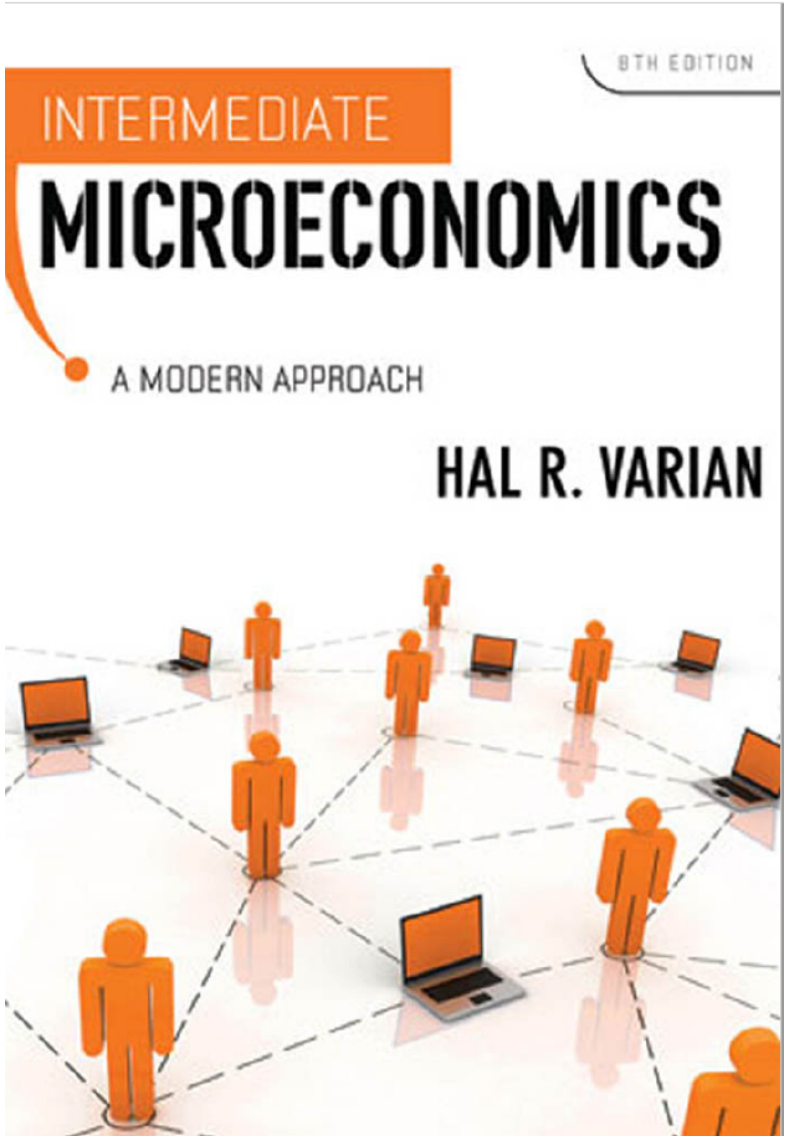
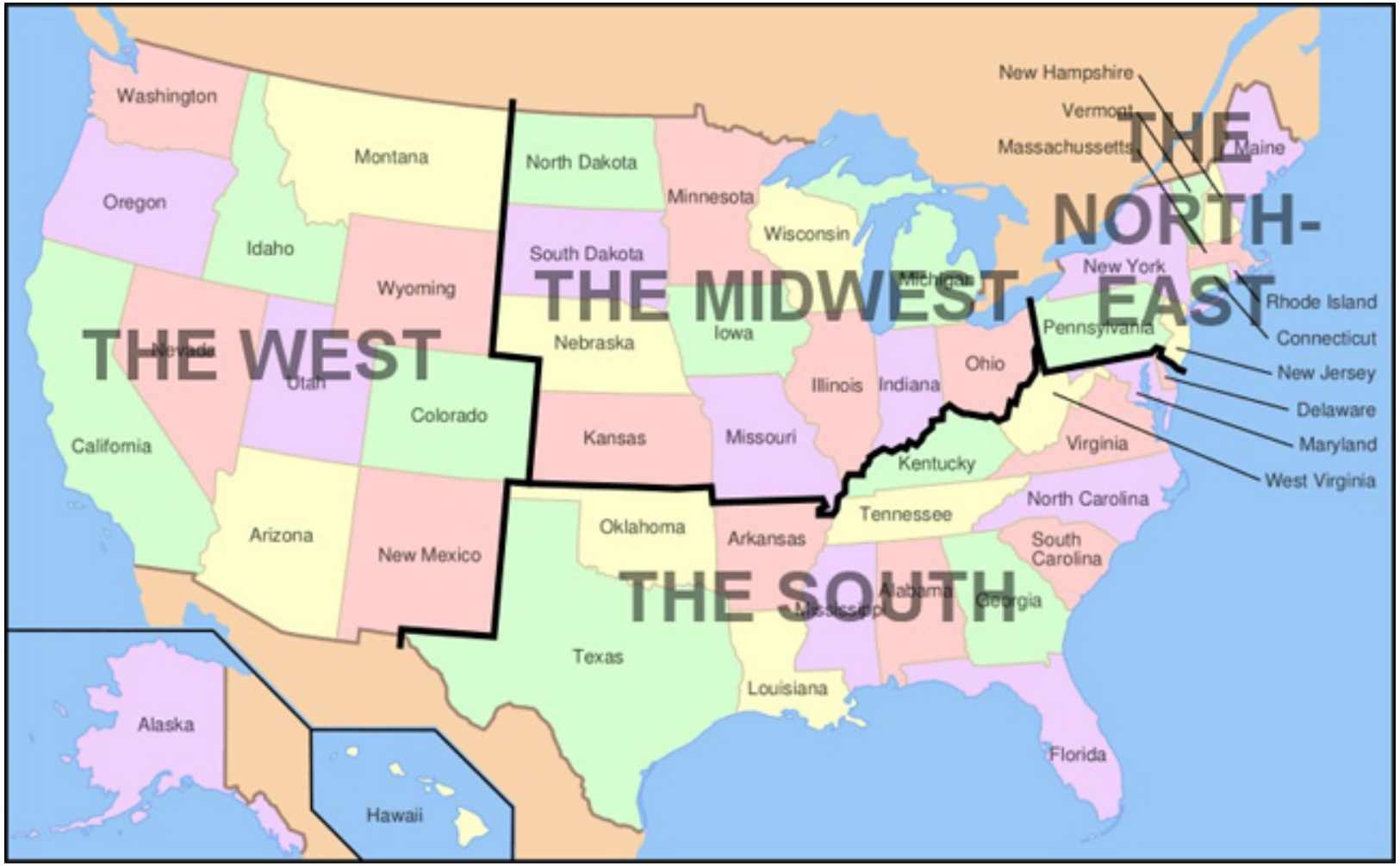

8TH EDITION
INTERMEDIATE
MICROECONOMICS
A MODERN APPROACH
HAL R. VARIAN

Constructing a Model

- Economics: Developing models of social phenomena.
- We will start with an example: market for apartments in a medium sized mid-western college town.
- Types of variables involved:
  - Endogenous variables
  - Exogenous variables

Regions in the USA



**Source: North America Conferences

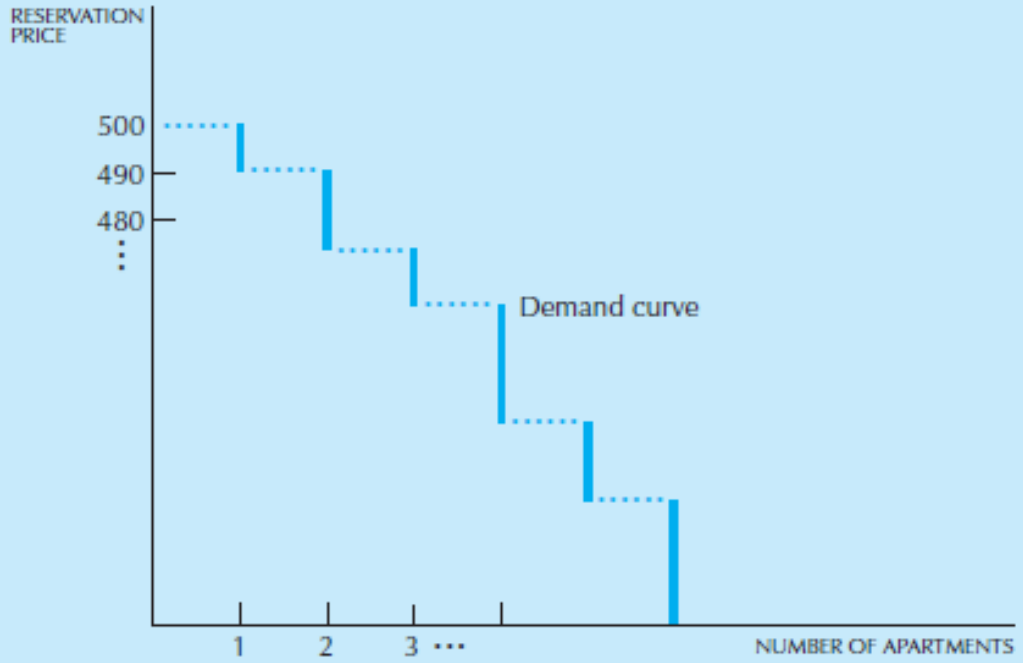
Optimization and Equilibrium

- *The optimization principle:* People try to choose the best patterns of consumption that they can afford
- *The equilibrium principle:* Prices adjust until the amount that people demand of something is equal to the amount that is supplied.

The Demand Curve

- Consider all possible renters of apartments.
- List the highest amount each is willing to pay.
- A person's maximum **willingness to pay (WTP)** is called the person's **reservation price**.
- Let us plot the reservation prices of every individual.

The Demand Curve



The demand curve for apartments. The vertical axis measures the market price and the horizontal axis measures how many apartments will be rented at each price.

Figure 1.1

**Source: Intermediate Microeconomics by Hal R. Varian

The Demand Curve

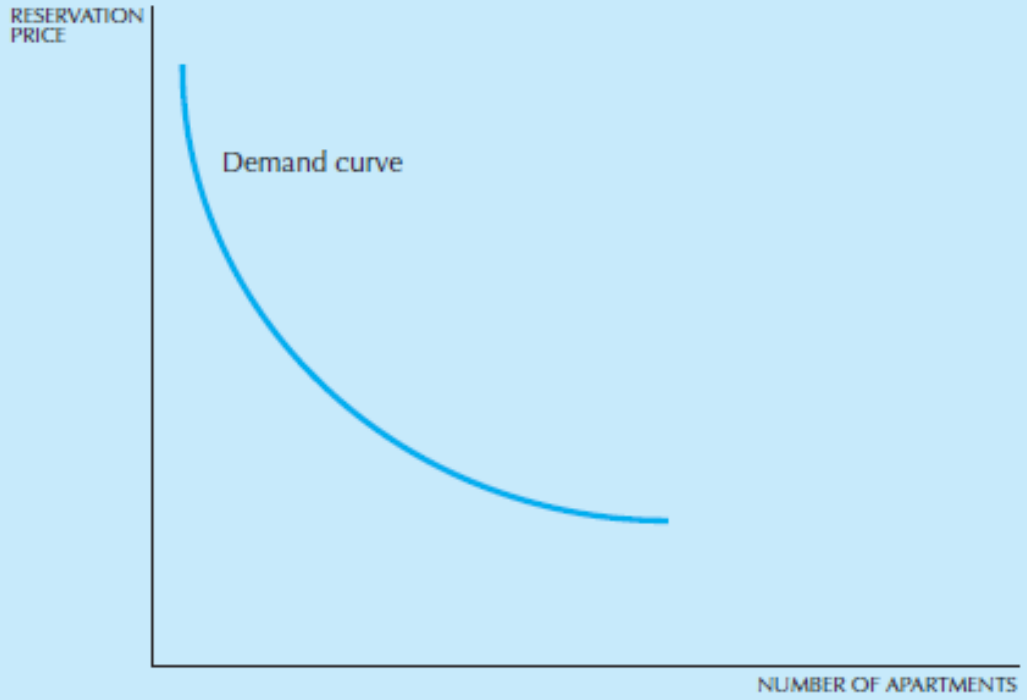


Figure 1.2 **Demand curve for apartments with many demanders.** Because of the large number of demanders, the jumps between prices will be small, and the demand curve will have the conventional smooth shape.

**Source: Intermediate Microeconomics by Hal R. Varian

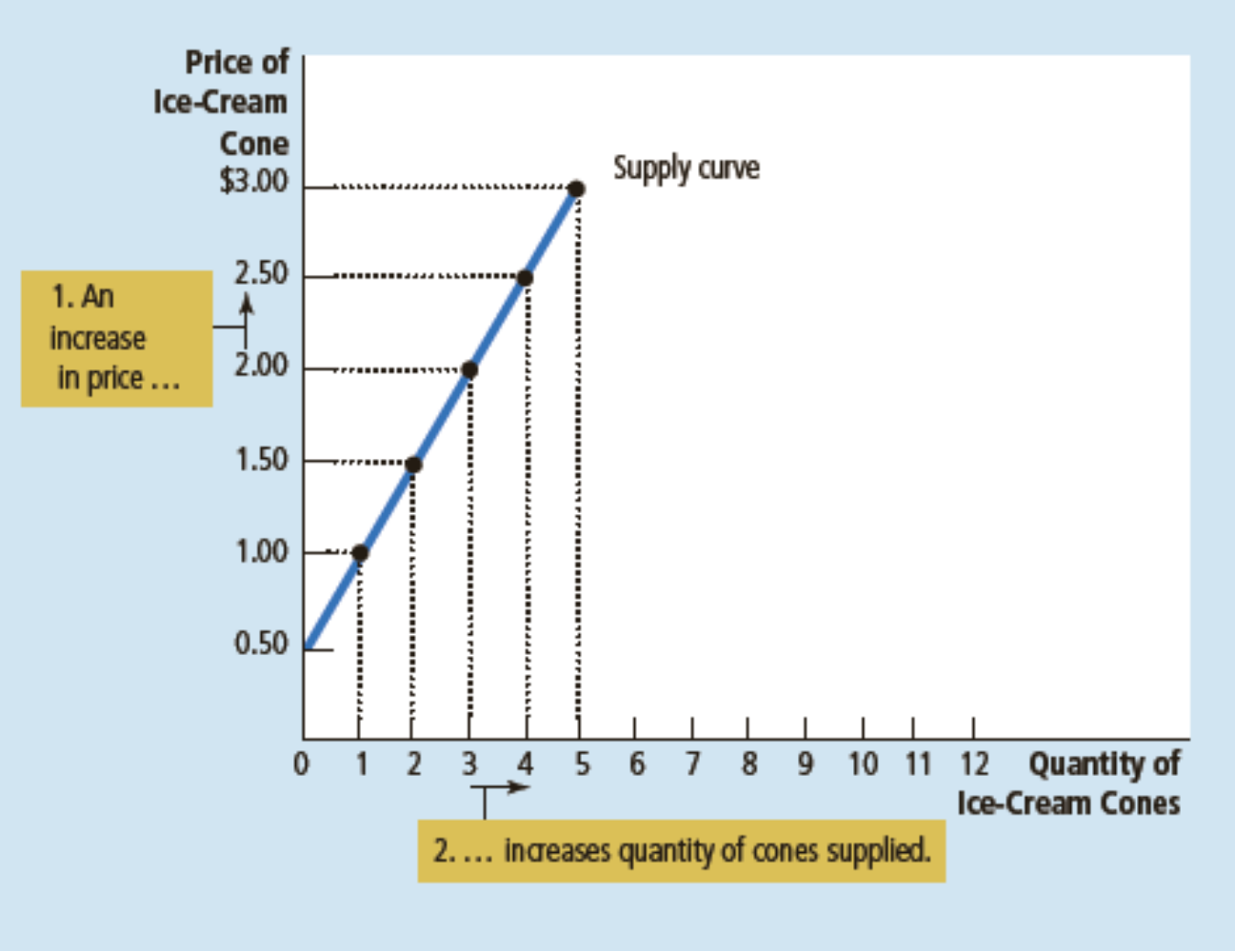
Supply

- Quantity supplied is a sort of mirror of demand, but in a sense, is easier to understand.
- It is the minimum a seller is willing to accept (WTA).
- It's easy to guess how this will reflect in a two dimensional curve.
- Let's understand the idea in detail.

Aggregate and Exogenous Factors

- Supply curves can be aggregated.
- Exogenous variables include
  - Input Prices.
  - Technology (costs).
  - Expectations (A little lesst sraightforward).

The Supple Curve



**Source: Micro by Mankiw (8e)

The Supply Curve: Let's Keep it Siple for now.

- Competitive market.
- Equilibrium price must be the same
- Short run supply is fixed.
- You will see that generally, supply is upward sloping in future chapters. However, in this case, it is fixed.
- This is because, in the short run, the number of apartments cannot change.

The Supply Curve

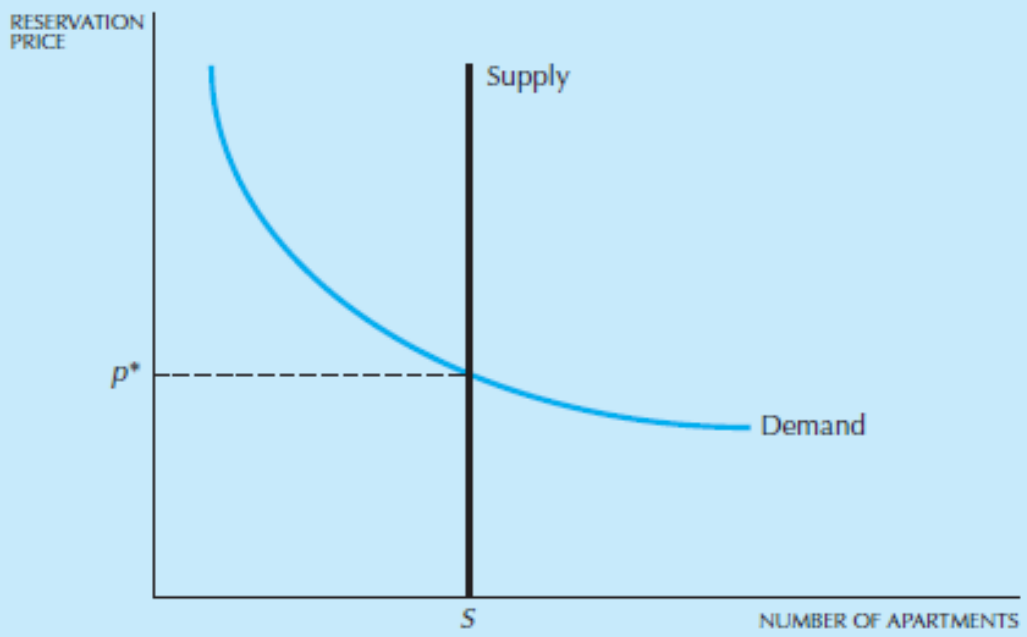


Figure 1.4 **Equilibrium in the apartment market.** The equilibrium price, p^*, is determined by the intersection of the supply and demand curves.

**Source: Intermediate Microeconomics by Hal R. Varian

Market Equilibrium

- Represented by point of intersection between demand and supply.
- Everyone willing to pay the equilibrium price gets an apartment, and the market clears.
- Others choose an alternate option.

Market Equilibrium

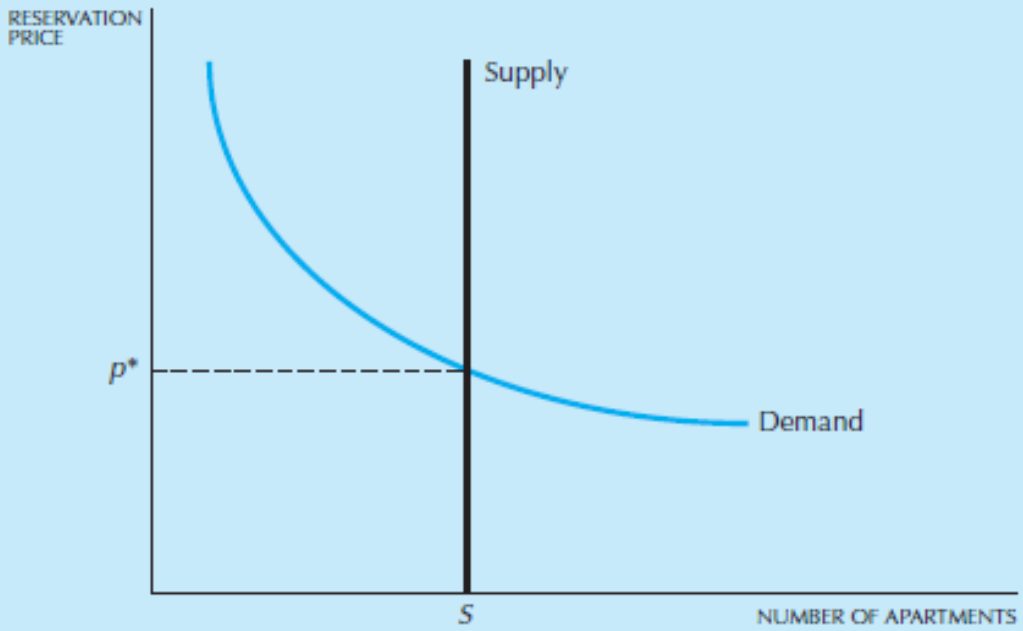


Figure 1.4 **Equilibrium in the apartment market.** The equilibrium price, p^*, is determined by the intersection of the supply and demand curves.

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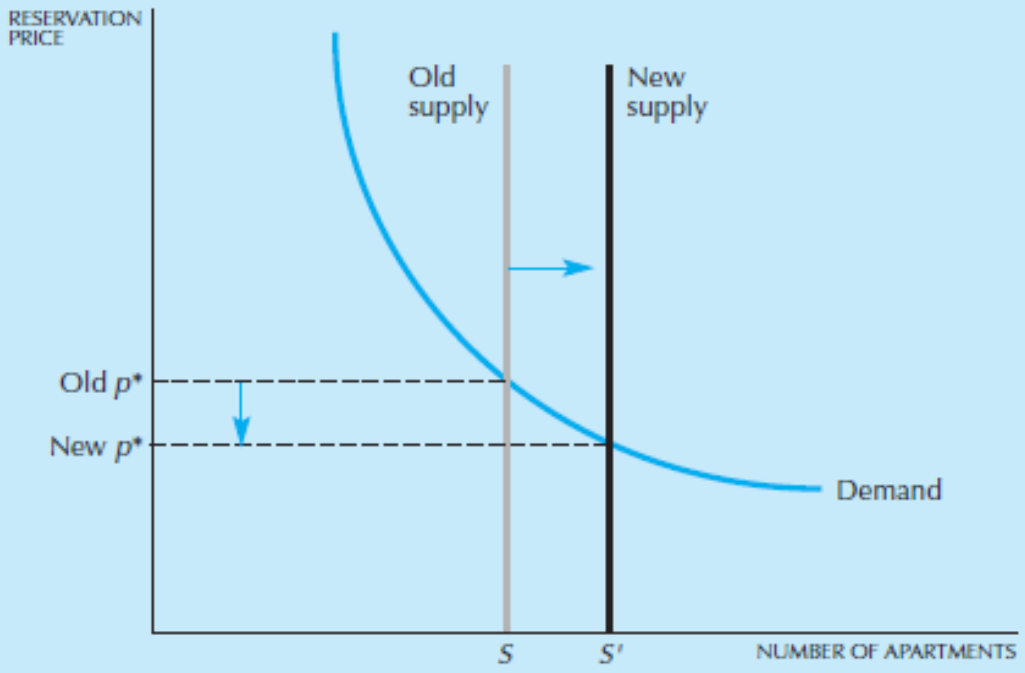
Comparative Statics

- Change in equilibrium as certain exogenous variables change.
- Analysis of *shifts* in demand and supply.
- Primarily concerned with comparing equilibria.

Comparative Statics



Increasing the supply of apartments. As the supply of apartments increases, the equilibrium price decreases.

Figure 1.5

**Source: Intermediate Microeconomics by Hal R. Varian

Comparative Statics

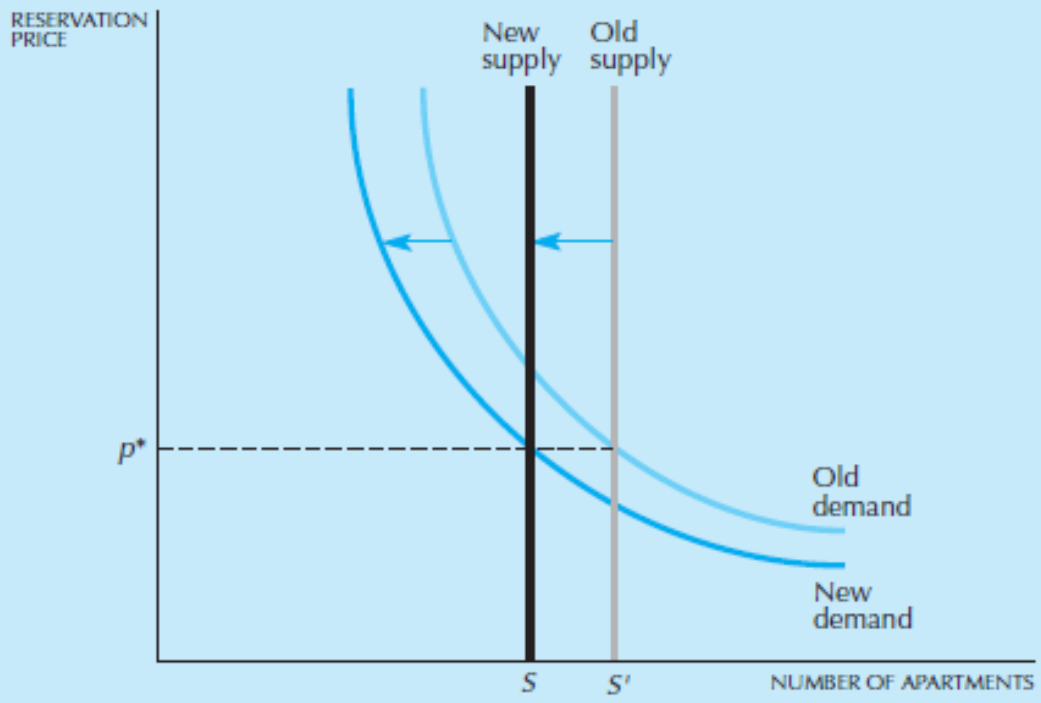


Figure 1.6 **Effect of creating condominiums.** If demand and supply both shift left by the same amount the equilibrium price is unchanged.

**Source: Intermediate Microeconomics by Hal R. Varian

Other ways to Allocate Apartments

The discriminating monopolist

- Auction them off one by one.
- Different people end up paying different prices, and everyone pays their maximum willingness to pay.
- Same people receive the apartment as in a competitive setup, but prices are different.

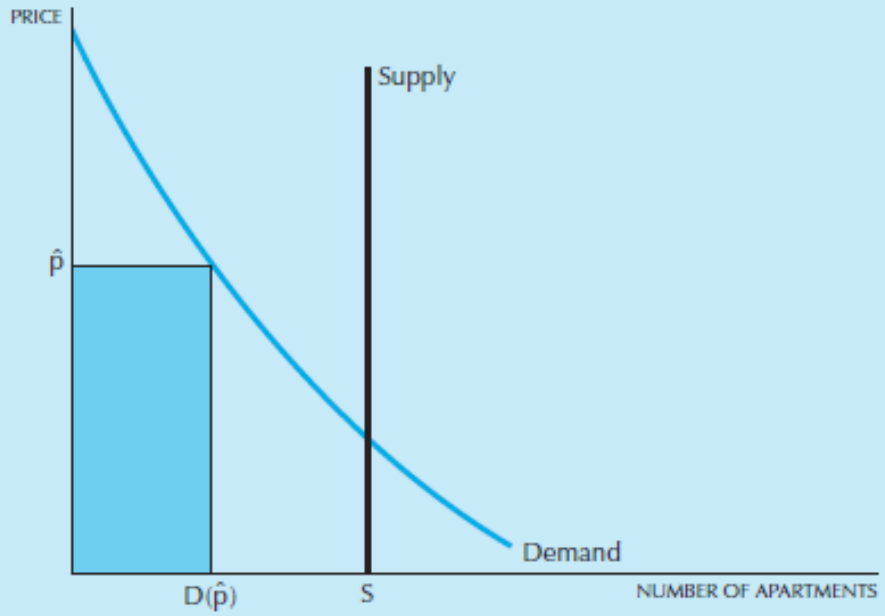
Other ways to Allocate Apartments

The ordinary monopolist

- One landlord who owns all the apartments.
- If the monopolist has to choose a single price p, he will try to maximize his revenue $pD(p)$.
- There is a trade off between the price of apartments and the number of apartments sold.
- The optimal revenue is obtained at $\hat{p}$.
- This assumes 0 cost of selling the apartment.

Other ways to Allocate Apartments

The ordinary monopolist



Revenue box. The revenue received by the monopolist is just the price times the quantity, which can be interpreted as the area of the box illustrated.

Figure 1.7

**Source: Intermediate Microeconomics by Hal R. Varian

Pareto Efficiency

- *Pareto improvement*: When at least one agent is made better off without any agent being made worse off.
- *Pareto Efficient*: No one can be made better off without at least one agent being made worse off.
- *Pareto inefficient*: When Pareto improvement is possible.

Other Topics

- Comparing ways to allocate apartments.
- Equilibrium in the long-run.

Supply

- We will see how demand and supply interact in various cases.
- We jump to a slightly general case here, where we assume supply to be upward sloping.
- We will discuss why in future chapters in more detail.

Market Equilibrium

- We rely on aggregate demand and aggregate supply.
- This is basically a sum of all individual demand and supply curves.
- Note that quantities get added here, while prices get averaged.

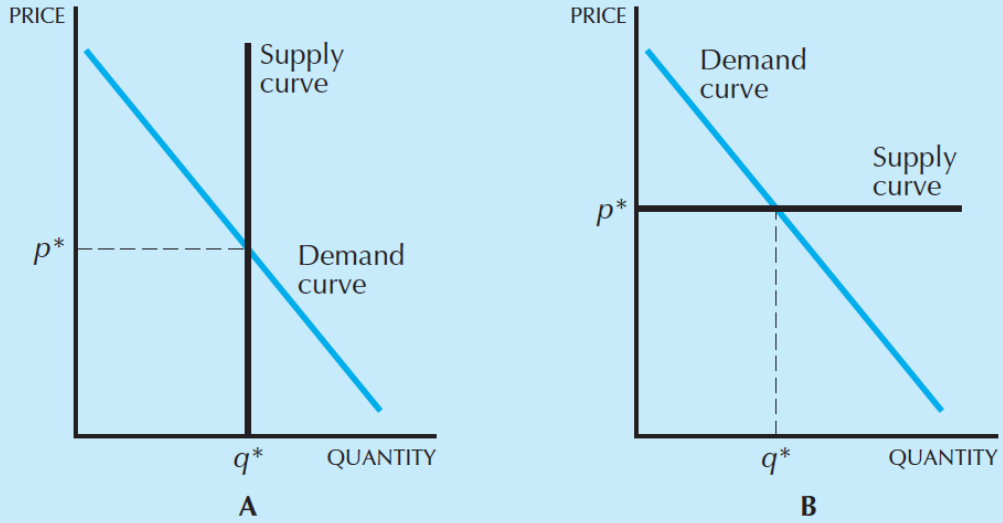
Market Equilibrium

- Both supply and demand are functions of price.
- In equilibrium, when $p = p^*$, we have

$$D(p^*) = S(p^*)$$

- Geometrically, this is when the demand and supply cross each other.

Two Special Cases



Special cases of equilibrium. Case A shows a vertical supply curve where the equilibrium price is determined solely by the demand curve. Case B depicts a horizontal supply curve where the equilibrium price is determined solely by the supply curve.

Figure 16.1

**Source: Intermediate Microeconomics by Hal R. Varian

Inverse Demand and Supply

- The inverse demand curve is given by $P_S(q^*) = P_D(q^*)$.
- Consider the following examples of linear demand and supply

$$D(p) = a - bp \qquad S(p) = c + dp$$

- In equilibrium

$$D(p) = a - bp = c + dp = S(p)$$

Inverse Demand and Supply

- Solving, we get

$$p^* \equiv \frac{a-c}{d+b}$$

- Substituting this into $D(p^*) = a - bp^*$,

$$D(p^*) = \frac{ad+bc}{b+d}$$

Inverse Demand and Supply

- Inverse demand is given by $P_D(q)$ and $P_S(q)$.
- Since $q = a - bp$,

$$P_D(q) = \frac{a - q}{b}$$

- Similarly,

$$P_S(q) = \frac{q - c}{d}$$

Inverse Demand and Supply

- We now have

$$P_D(q) = \frac{a-q}{b} = \frac{q-c}{d} = P_S(q)$$

- Which yields

$$q^* = \frac{ad+bc}{b+d}$$

Comparative Statics

$$D(p) - m \qquad S(p) - m$$

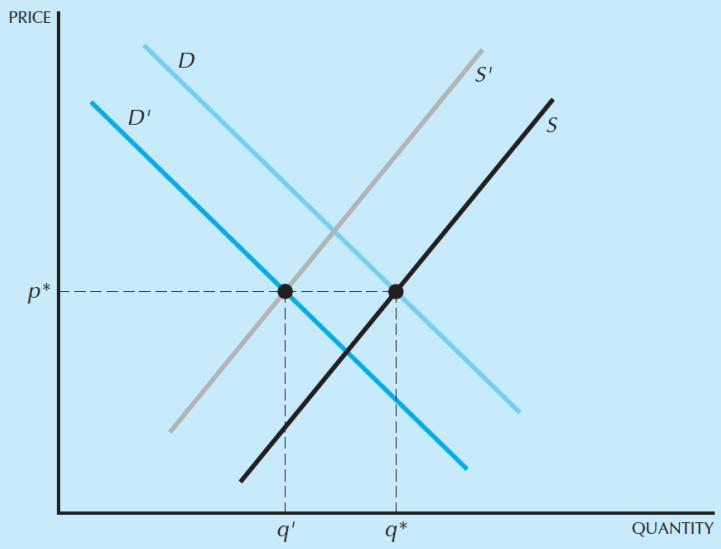


Figure 16.2 **Shifting both curves.** Both demand and supply curves shift to the left by the same amount, which implies the equilibrium price will remain unchanged.

**Source: Intermediate Microeconomics by Hal R. Varian

Taxes

- Quantity taxes:
  - $P_D = P_S + t$.
- Value or ad valorem taxes:
  - $P_D = (1 + \tau)P_S$.
- In the linear case, at the intersection of demand and supply, we have,
  - $a - bp_D = c + dp_S$.
  - $p_D = p_S + t$.

Taxes

- Substituting p_D into the above equation yields

$$a - b(p_S + t) = c + dp_S$$

- Solving yields

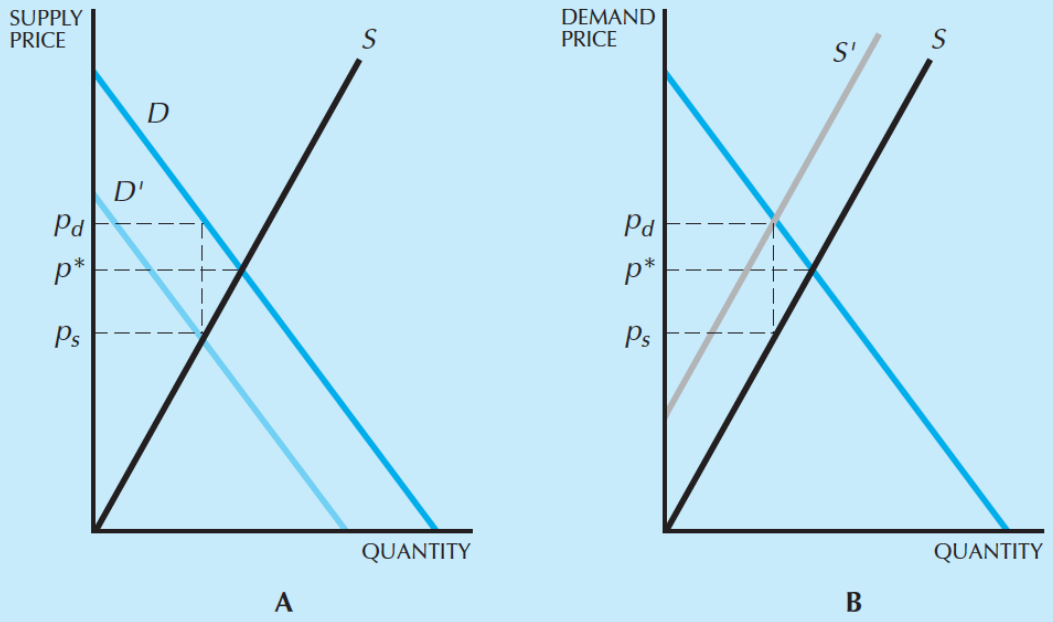
$$p_S^* = \frac{a - c - bt}{d + b}$$

- Thus, p_D^* is given by

$$p_D^* = \frac{a - c - bt}{d + b} + t$$

$$\Rightarrow p_D^* = \frac{a - c + dt}{d + b}$$

Taxes



The imposition of a tax. In order to study the impact of a tax, we can either shift the demand curve down, as in panel A, or shift the supply curve up, as in panel B. The equilibrium prices paid by the demanders and received by the suppliers will be the same either way.

Figure 16.3

**Source: Intermediate Microeconomics by Hal R. Varian

Taxes

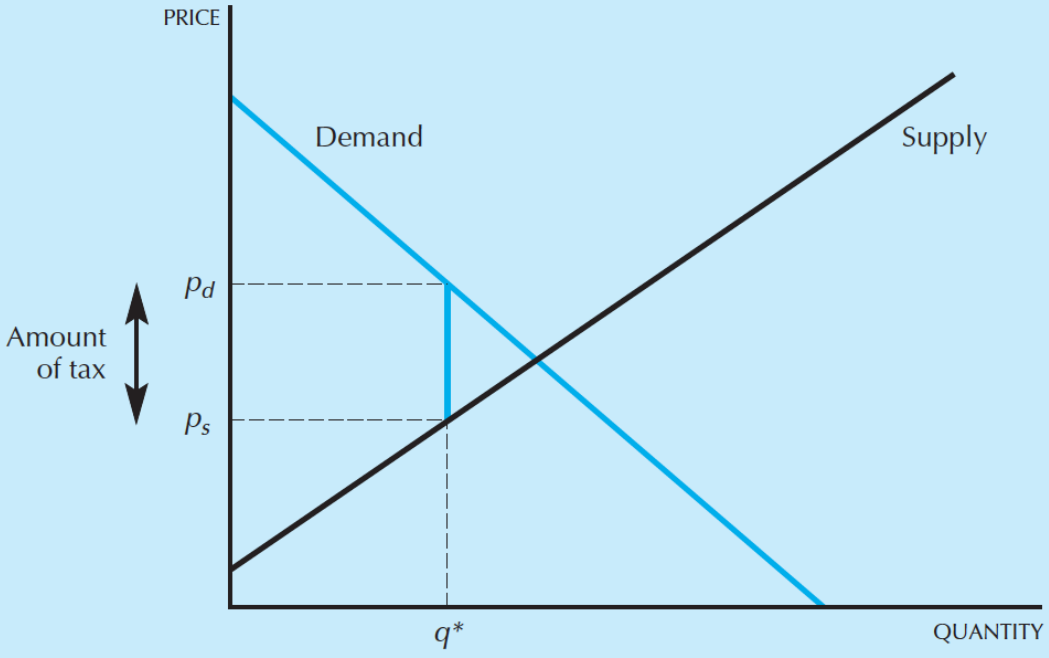


Figure 16.4 **Another way to determine the impact of a tax.** Slide the line segment along the supply curve until it hits the demand curve.

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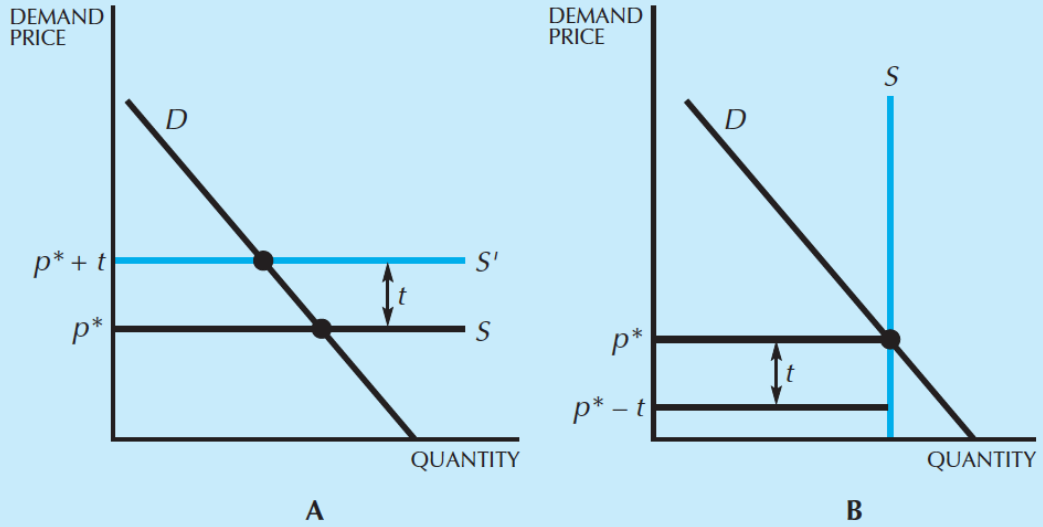
Passing along a Tax

- Tax on producers often doesn't hurt profits.
- Depends on the nature of demand and supply of these goods.

Taxes



Special cases of taxation. (A) In the case of a perfectly elastic supply curve the tax gets completely passed along to the consumers. (B) In the case of a perfectly inelastic supply none of the tax gets passed along.

Figure 16.5

**Source: Intermediate Microeconomics by Hal R. Varian